

Predicting Postoperative Atrial Fibrillation Following Cardiac Surgery Using Postoperative Atrial Fibrillation Score

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ABSTRACT

Postoperative atrial fibrillation (POAF) is one of the most common complications following heart surgery. Postoperative atrial fibrillation has severe clinical and financial consequences for patients' outcomes. Patients who develop postoperative atrial fibrillation have a higher risk of hospital readmission, in-hospital morbidity and mortality, and one-year mortality. Risk assessment and tailored treatment for high-risk patients may be vital in reducing the morbidity, mortality, and cost burden associated with this cardiac arrhythmia. Risk stratification models like Postoperative Atrial Fibrillation (POAF) Score have been developed and adapted to identify patients with an increased likelihood of developing postoperative atrial fibrillation (POAF) in order to institute preventive measures in the preoperative and early postoperative period. The study assessed the effectiveness of Postoperative Atrial Fibrillation Score (POAF Score) in predicting the occurrence of POAF. This study is a cross-sectional study on consecutive patients that had heart valve surgeries and coronary artery bypass grafting at the National Cardiothoracic Centre, Accra, Ghana, over a one-year period. The POAF Score risk assessment tool was assessed in the perioperative period. The patients were monitored in the postoperative period for the development of postoperative atrial fibrillation in the first five postoperative days. The two were tested for any significant association. Twenty-four patients with mean age of 47.75 years $\pm$ 16.63 were studied, and the incidence of postoperative atrial fibrillation was found to be 29% in the study population, mostly occurring on postoperative day 2 and 3. Using the receiver operating characteristic (ROC) curve, the predictability of POAF score was found to be 29.41%. Postoperative Atrial Fibrillation Score does not accurately predict the occurrence of postoperative atrial fibrillation in this study population.

Keywords: Postoperative atrial fibrillation, postoperative atrial fibrillation score, atrial fibrillation, cardiac surgery.

INTRODUCTION

Atrial fibrillation is an atrial tachyarrhythmia characterized by predominantly uncoordinated atrial activation with consequent deterioration of atrial mechanical function.¹ Electrocardiographic finding consistent with atrial fibrillation includes the absence of a consistent p-wave; rapid oscillations or fibrillatory waves that vary in shape, size, and timing; and an irregular ventricular response when atrioventricular conduction is intact.¹ Postoperative atrial fibrillation (POAF) has been defined as the occurrence of atrial fibrillation lasting more than 5-10 minutes following surgery.² Postoperative atrial fibrillation is linked to haemodynamic instability, acute embolic

stroke, and organ dysfunction from embolic events (particularly in individuals with left ventricular dysfunction and heart failure).³

The pathophysiology of postoperative atrial fibrillation is not well understood, but certain factors such as atrial dilatation, age-related fibrosis of the heart, structural damage to the heart, hypertension, diabetes mellitus, left ventricular hypertrophy, chronic obstructive pulmonary disease, and sepsis have been identified as contributors to the development of this tachyarrhythmia.^{4,5}

The incidence of postoperative atrial fibrillation following noncardiac surgery is about 5-10% and as high as 60% following cardiac surgery.⁶ The highest incidences have been seen in patients undergoing combined valve surgery and coronary artery bypass grafting (CABG) followed by isolated CABG, heart transplantation, and noncardiac surgeries, in descending order.⁴ It typically occurs on the second postoperative day and is usually transient but usually reoccurs in up to 40%

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of patients.⁷ Postoperative atrial fibrillation is associated with increased postoperative complications and a more extended hospital stay (additional 4-5 days).^{4,8} The incidence of hypotension, pulmonary oedema, cerebrovascular accidents, renal dysfunction, pneumonia, deep surgical site infection, vascular catheter site infection, a decline in mental state score, and genitourinary infections are higher in patients who develop postoperative POAF compared to patients who do not develop POAF (22.6% vs 15.4%).⁸ Acute limb ischaemia and the need for reoperation have also been seen to increase in patients that develop POAF. Postoperative atrial fibrillation has also been associated with lower in-hospital and long-term survival; this observation has been noticed up to four years after surgery.⁹ In-hospital mortality was about 4.7% in patients that developed POAF after CABG compared to 2.1% in those that did not develop POAF in a multicenter study involving 4,657 patients carried out by Mathew PJ *et al.*¹⁰ In the United States, the cost of care for patients who developed postoperative atrial fibrillation was increased by approximately \$10,000 to \$11,500 per patient.^{4,11}

Predicting postoperative atrial fibrillation will directly impact the development and institution of preventive measures for patients with a high risk of developing POAF. These measures will reduce postoperative morbidity, length of hospital stay, and mortality following cardiac surgery. In addition, this may also reduce the financial burden on the patient and reduce the burden on the already inadequate hospital facilities and workforce in our environment. Risk stratification models have been developed and adapted to identify patients with an increased likelihood of developing POAF. These include Postoperative Atrial Fibrillation Score, Atrial Fibrillation Risk Index and Congestive Heart Failure, Hypertension, Age, Diabetes, Stroke, Vascular Disease, Age, Sex (CHA2DS2VASc) Score.¹²

This study assessed the effectiveness of Postoperative Atrial Fibrillation (POAF) score in predicting the occurrence of postoperative atrial fibrillation in adult patients that had valve replacement surgery and coronary artery bypass grafting surgery.

MATERIALS AND METHODS

Study Population and Design

This is a cross-sectional study that sought to ascertain the accuracy of postoperative atrial fibrillation score in predicting postoperative atrial fibrillation in 24 patients that had heart valve and coronary artery bypass graft surgeries at the National Cardiothoracic Centre, Korle-Bu Teaching Hospital Accra, Ghana between November 2021 and October 2022. Patients with established atrial fibrillation, patients on medications for the prevention of atrial fibrillation and those who refused to give consent to be included in the study were excluded from this study.

Study Area

The study was carried out at the National Cardiothoracic Centre, Korle-Bu Teaching Hospital, Accra, Ghana, a referral centre for most West African countries that carries out a multitude of surgeries in both adult and pediatric cardiac and thoracic surgery. An average of 20 valve surgeries and three coronary artery bypass graft surgeries are performed annually.¹³

Patient Evaluation

Patients on admission for heart valve surgeries and CABG at the National Cardiothoracic Centre who fulfilled the inclusion criteria had the study's objectives explained, and an informed consent was obtained. The demographic characteristics of the patients, relevant history (especially drug history), and physical examination findings (especially the presence of atrial fibrillation) were recorded on the Proforma. A preoperative electrocardiogram was done as part of the preoperative work up.

The postoperative atrial fibrillation score was calculated for each patient preoperatively. The score consists of seven parameters that were used to predict the

occurrence of postoperative atrial fibrillation. These include.¹² :

	Parameter	Years	Score
1	Age	60-69 years	1
		70-79 years	2
		80 years and above	3
2.	Chronic Obstructive Pulmonary Disease		1
3.	Estimated Glomerular Filtration Rate <15mL/minute or Dialysis		1
4.	Emergency surgery		1
5.	Preoperative intra-aortic balloon pump		1
6.	Left ventricular ejection fraction < 30%		1
7.	Valve surgery		1

Patients with POAF scores < 3 were considered low risk for developing postoperative atrial fibrillation, while those

with scores of 3 and above were considered high risk for developing postoperative atrial fibrillation.¹¹

Intraoperative Details

Surgical access for all patients was through a median sternotomy. All the procedures were done under cardiopulmonary bypass with either aorto-bicaval or a two-stage venous cannulation. The procedures were done in a standard fashion, and the patients were weaned off cardiopulmonary by-pass in the routine fashion with minimal inotropic requirement. They were transferred to the intensive care unit on mechanical ventilation.

Postoperative Monitoring

Patients were weaned off mechanical ventilation when fully awake, had reasonable respiratory effort, good haemodynamic profile and acceptable arterial blood gases. Patients were evaluated daily, and features of atrial fibrillation were explicitly looked for (palpitation, syncope, irregularly irregular heart rhythm). Continuous electrocardiographic monitoring was instituted and recorded on the patient's ICU chart, and a standard 12-lead electrocardiogram was obtained daily for the first five days following surgery using a

Philips Page Writer TC20 Cardiograph. A diagnosis of postoperative atrial fibrillation was made when there was a new onset of an irregularly irregular pulse rate associated with electrocardiographic findings of an absent p-wave and an irregularly irregular rhythm in the postoperative period.⁴

Statistical Analysis

Data obtained was imputed into a computer and analyzed using the statistical software for social sciences (SPSS) version 25. Quantitative variables (such as age, BMI, heart rate, et cetera.) were presented as mean and standard deviation, while categorical variables (such as gender, heart rhythm, et cetera.) were presented as frequencies and percentages.

Pearson's Chi-square test was used to identify the association between the postoperative atrial fibrillation (POAF) score categories and the presence of atrial fibrillation. The Receiver Operating Characteristic (ROC) curve was used to determine whether or not the POAF score predicts the presence of atrial fibrillation. The significance level was $p < 0.05$ at a 95%

confidence level. Ethical clearance was obtained from the Institutional Review Board of the Korle-Bu Teaching Hospital, Accra, Ghana. Permission was also sought and granted by the National Cardiothoracic Centre, Accra.

RESULTS

A total of 24 patients were recruited for the study within the study period. The age range was between 17 and 70 years. The mean age was 47.75 ± 16.63 (standard deviation). The mean body mass index (BMI) was $26.20 \pm 8.20 \text{ kg/m}^2$ (Table 1). Fifty per cent of the patients (12) were male. Eight patients had mitral valve replacement, while only one had combined mitral valve replacement and coronary artery bypass grafting (Table 2).

Table 1: Sociodemographic Characteristics

Variables	Minimum	Maximum	Mean	Std. Deviation
Age (Years)	17.00	70.00	47.75	16.63
Height (cm)	107.00	193.50	164.90	15.74
Weight (kg)	37.00	102.00	69.73	16.80
BMI (kg/m ²)	13.40	55.00	26.20	8.20

Abbreviation: BMI; Body Mass Index, STD; Standard

Seven patients (29%) were found to have atrial fibrillation postoperatively. Only one patient was found to have a high risk of developing postoperative atrial fibrillation using the postoperative atrial fibrillation score (POAF). The patient did not develop atrial fibrillation in the postoperative period. Using the receiver operating characteristic (ROC) curve, the POAF score's predictability was 29.41% (Figure 1).

Using logistic regression, it was found that patients between 31-40 years of age are three times more likely to develop POAF than those between 18-30. This was not found to be statistically significant ($p=0.105$). Patients between 61-70 years of

age were two times more likely to develop POAF when compared to those between 18-30 years. This was, however, not found to be statistically significant ($p=0.053$) (Table 3). Both gender and body mass index were not found to correlate significantly with the development of POAF (Table 3).

DISCUSSION

Postoperative atrial fibrillation significantly contributes to postoperative morbidity and mortality in adults following cardiac surgery. This study investigated the ability of POAF score to predict its occurrence in adult patients that are undergoing coronary artery bypass grafting and valve surgeries.

The mean age of the patients in this study (47.75 ± 16.63 years) is similar to a study by Lung *et al.*,¹⁴ which found the median age of development of valvular heart disease (accounting for about 92% of cases in this study) to be 43 years. It is, however, at variance with the results of studies conducted in the developed world, which revealed the mean age to be 75 years.¹⁴⁻¹⁶ The reason for this variation is possibly related to the etiology of valvular heart disease in the different parts of the world, commonly rheumatic heart disease in developing countries like Ghana, and age-related sclerosis in the developed world.

The incidence of POAF in this study (29%) is similar to the findings of Leloir *et al.*¹⁷ and Mariscalco *et al.*¹¹, where the incidence of POAF following cardiac surgery was found to be 26.7% and 26.4%, respectively. Other researchers found the incidence to be as high as 40% (ranging from 25-60% depending on the surgical procedure)^{8,9,18,19}, which is at variance with this study. The difference may be attributed to the relatively lower mean age of patients in this study compared the other studies^{9,19} (47.75 ± 16.63 versus 63.8 ± 13.1 and 65.5 ± 12) because increasing age is a significant contributor to the development of POAF. The increasing age of the study population has

been linked to developing all forms of atrial fibrillation, including POAF.

Table 2: Preoperative Details

Variable	Total n(%)	Sinus n(%)	Atrial Fibrillation n(%)
Gender			
Female	12(100.0)	10(83.3)	2(16.2)
Male	12(100.0)	7(58.3)	5(41.7)
Procedure			
AVR	7(100.0)	5(71.4)	2(28.6)
BENTALL	1(100.0)	1(100.0)	0(0.0)
CABG	2(100.0)	1(50.0)	1(50.0)
CABG+MVR	1(100.0)	1(100.0)	0(0.0)
DVR	4(100.0)	3(75.0)	1(25.0)
MVR	8(100.0)	5(62.5)	3(37.5)
TVR	1(100.0)	1(100.0)	0(0.0)

Abbreviations; AVR Aortic valve replacement, CABG Coronary artery bypass grafting, MVR Mitral valve replacement, DVR Double valve replacement, TVR Tricuspid valve replacement

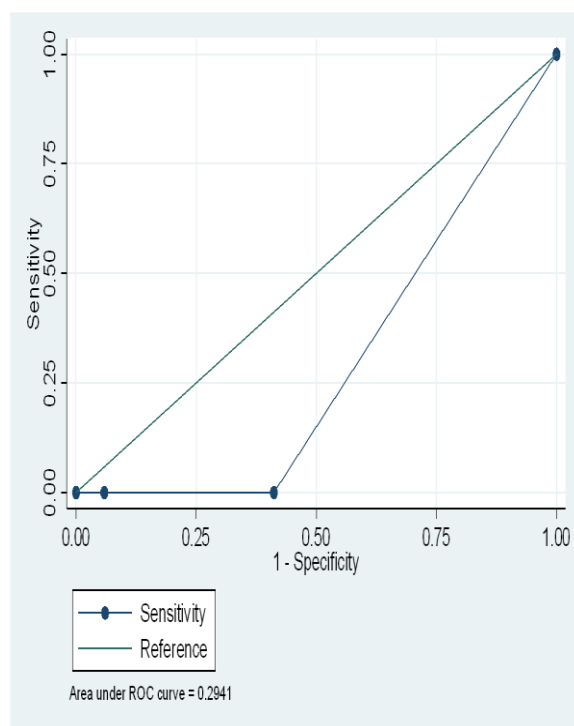


Figure 1: ROC curve showing predictability of POAF score

Most patients that developed POAF did so within the second and third postoperative days, a similar finding to that of Jongnarangsin et al.⁸ and Haffajee et al.⁹ The reason for the occurrence during the first five postoperative days has not been

extensively studied. However, it may be related to the peak of inflammation occurring in the postoperative period as seen by a rise in the inflammatory markers, including CRP, Tumour Necrosis Factor-Alpha (TNF- α), Interleukin-6 (IL-6), renin-angiotensin system and White blood cell count (WBC)²⁰. This forms the basis for using certain medications like methylprednisolone, angiotensin-converting enzyme inhibitors and vitamin C to limit inflammatory response and prevent POAF.

The low predictability of the POAF score in this study (0.29 (95% CI, 0.17 to 0.41)) is at variance with the findings of Mariscalco et al.¹¹ and Cameron et al.¹² who had a predictability of 0.64 and 0.65 respectively. The likely reason may be due to the low sample size in this study compared to the other studies with several thousand patients. It may also be because the patients in this study have a different risk profile than those in the other studies, with only one patient having a POAF score of 3 and above. In this study, the majority were middle-aged, none had COPD, none had emergency surgery, only one patient had a low ejection fraction (EF), and all had good kidney functions (eGFR) prior to surgery. These led to a low POAF score among this study population. The POAF score has not been validated in this highly selected patient population.

Preoperative clinical characteristics were also evaluated and compared between those that developed POAF and those that did not. The finding that advancing age was associated with an increased likelihood of developing postoperative atrial fibrillation is similar to several studies that have also found that advancing age is an independent predictor of POAF.^{17,21,22} Age-related myocardial degeneration, dilatation and nonuniform conduction are thought to be associated with the development of atrial fibrillation in the general population.⁸ Female gender was found to be four times more likely to develop POAF, but this was not statistically significant. This is at variance with the study by Smith et al.²²,

Leloir et al.¹⁷, Jongnarangsin et al.¹⁸ and Staerk et al.²¹, who found that the male gender is an independent risk factor for the development of POAF. This may be related to the epidemiology of coronary artery disease, which has been traditionally more common in the male gender. The reason for the variance in this study is not very clear and may need to be studied further in a larger study population. Also, obese patients were found to be slightly more likely to develop POAF (1.5 times), but this was also not statistically significant. Some researchers have linked obesity and elevated body mass index (BMI) to known risk factors of atrial fibrillation like Diabetes, myocardial infarction, left ventricular hypertrophy, left atrial enlargement, and Hypertension.²¹ They have also been independently linked to the development of atrial fibrillation.²¹

Table 3: Clinical characteristics and postoperative heart rhythm

Variable	Total n(%)	Sinus n(%)	Atrial Fibrillation n (%)	OR (95%CI, I)	P value
Age					0.064
18-3	3(100.0)	1(33.3)	2(66.7)	1	
31-40	6(100.0)	5(83.3)	1(16.7)	3.12(0.57-5.56)	0.105
41-50	4(100.0)	1(25.0)	3(75.0)	1.40(0.07-8.23)	0.826
51-60	3(100.0)	3(100.0)	0(0.0)	-	
61-70	8(100.0)	7(87.5)	1(12.5)	2.11(0.96-4.46)	0.053
Gender					0.85
Female	12(100.0)	10(83.3)	2(16.2)	3.57(0.53-23.95)	
Male	12(100.0)	7(58.3)	5(41.7)	1	
BMI					0.618
Under-weight	3(100.0)	1(33.3)	2(66.7)	1.10(0.05-8.91)	0.98
Normal	8(100.0)	7(87.5)	1(12.5)	1	
Over-Weight	6(100.0)	4(66.7)	2(33.3)	0.28(0.01-6.91)	0.44
Obese	7(100.0)	4(57.1)	3(42.9)	1.50(0.08-5.39)	0.77

Abbreviation; OR odds ratio, CI Confidence interval, BMI, body mass index

CONCLUSION

The incidence of POAF in the study population is 29%, mostly occurring between the second and third postoperative days. Postoperative Atrial Fibrillation Score is ineffective in predicting POAF in the study

population with a predictability of 29%. Advancing age is a statistically significant risk factor for the development of POAF.

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